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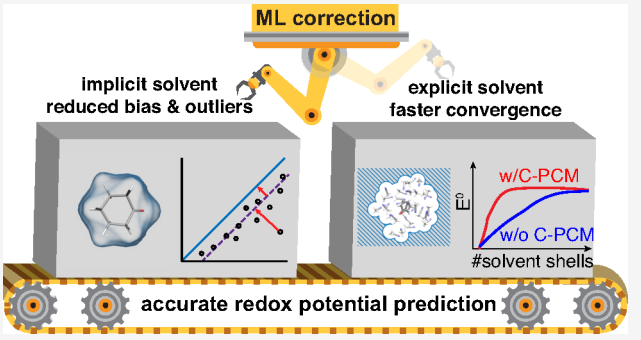
Bridging the Experiment-Calculation Divide: Machine Learning Corrections to Redox Potential Calculations in Implicit and Explicit Solvent Models

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ABSTRACT: Prediction of redox potentials is essential for catalysis and energy storage. Although density functional theory (DFT) calculations have enabled rapid redox potential predictions for numerous compounds, prominent errors persist compared to experimental measurements. In this work, we develop machine learning (ML) models to reduce the errors of redox potential calculations in both implicit and explicit solvent models. Training and testing of the ML correction models are based on the diverse ROP313 data set with experimental redox potentials measured for organic and organometallic compounds in a variety of solvents. For the implicit solvent approach, our ML models can reduce both the systematic bias and the number of outliers. ML corrected redox potentials also demonstrate less sensitivity to DFT functional choice. For the explicit solvent approach, we significantly reduce the computational costs by embedding the microsolvated cluster in implicit bulk solvent, obtaining converged redox potential results with a smaller solvation shell. This combined implicit−explicit solvent model, together with GPU-accelerated quantum chemistry methods, enabled rapid generation of a large data set of explicit-solvent-calculated redox potentials for 165 organic compounds, allowing detailed investigation of the error sources in explicit solvent redox potential calculations.



1. INTRODUCTION

The redox potential is a fundamental thermodynamic property that describes the tendency of a chemical species to lose or acquire electrons. It is essential in mechanism studies of catalysis[1,2] and energy storage.[3] Design and discovery in these fields needs rapid predictions of redox potentials of thousands or even millions of candidate compounds, which can hardly be achieved with cyclic voltammetry measurements. Recent developments in accelerated quantum mechanical (QM) methods and solvent models enabled rapid curation of computational redox potential data sets with thousands of molecules. Machine learning (ML) models trained on these computational data sets can accurately reproduce the QM calculated redox potentials, enabling the exploration of multimillion compound spaces for redox flow battery design[4] and biochemical discovery.[5]

However, the accuracy of QM predicted redox potentials relative to experiment strongly depends on the QM methods and solvent-model-related parameters. Highly accurate results (error ≤ 65 mV) are only found in small batch studies with specific combinations of methods and parameters applied to a specific class of compounds.[6] This uncertainty in QM calculations can impact the accuracy of curated computational redox potential data sets and, hence, the predictivity of ML models trained on these computational data sets used for design and discovery. Therefore, there is an urgent need to improve the accuracy of the QM prediction of redox potentials.

Due to the complexity of the solvent environment, errors in QM calculations of redox potentials can be attributed to many different sources. One prominent source is the solvation-free energy of species involved in the redox process. Most redox potential calculations were conducted in implicit solvent models, such as the conductor-like polarizable continuum models (COSMO,[7] C-PCM,[8−10] GCOSMO,[11] IEF-PCM[12−14]) and the variant COSMO-RS,[15] due to their computational efficiency. However, prominent errors have been observed in implicit solvent redox potential calculations, especially for highly charged compounds.[16,17] Systematic bias of the calculated redox potentials relative to the experiment is often corrected with a simple linear regression,[18] which, however, cannot deal with the different error sizes for differently charged species, nor can it remove large-error outliers. A few strategies have been proposed to correct the

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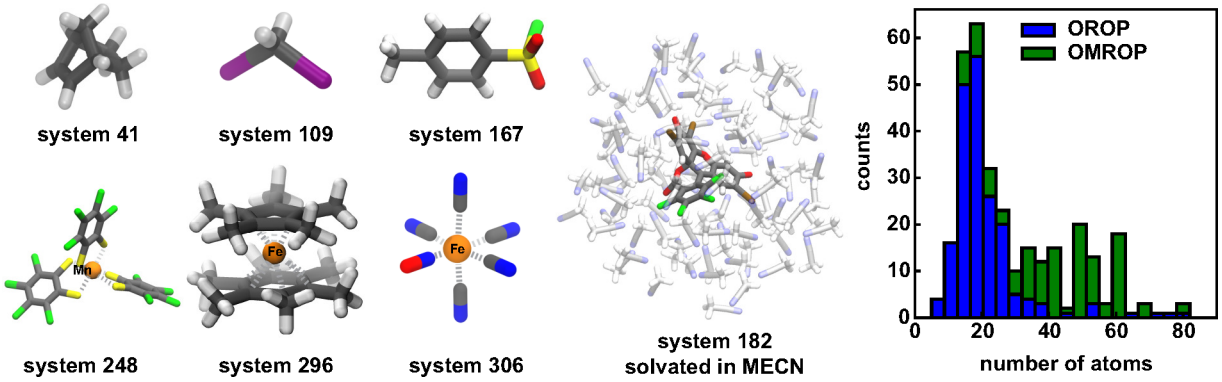


Figure 1. Representative systems from the ROP313 data set used in this work. (top left) Three representative organics including the cyclic aliphatic norbornane, the halocarbon diiodomethane, and the aromatic 4-methylbenzylsulfonyl chloride. (bottom left) Three representative organometallics with bidentate (3,4,5,6-tetrachlorobenzene-1,2-dithiolate), sandwich (pentamethylcyclopentadiene), and monodentate (CN^-, CO) ligands. (middle) Example structure of a microsolvated cluster used in our explicit solvent calculation. (right) The distribution of the number of atoms in each system. All atoms are colored by elements [C in gray, H in white, O in red, N in blue, Cl in green, S in yellow, I in purple, metal (Mn, Fe) in orange].

errors associated with highly charged species and verified to be effective on small data sets (ca. 20 compounds), such as the Pseudocounterion Solvation Scheme[16] and the variable-temperature H atom addition/abstraction.[17]

Explicit solvent approaches are shown to be more accurate for some challenging systems, such as transition metal complexes.[18−20] However, they can hardly be used in high-throughput computational design and discovery due to their significantly higher computational costs. The high computational costs are caused by the fact that converged redox potential results can only be obtained by including a large enough explicit solvent shell into QM calculations.[21,22] The optimal solvent shell size varies among different studies on different solute molecules.[23−25]

In this work, we develop ML models to reduce the errors of QM redox potential calculations in both implicit and explicit solvent models. Training and testing of the ML correction models are based on the large, diverse ROP313 data set[26] with experimental redox potentials of organic and organometallic compounds measured in different solvents. For the implicit solvent approach, we trained ML models to improve the accuracy relative to experiments across a diverse set of molecules regardless of the DFT functionals chosen. For the explicit solvent approach, we significantly reduce the computational costs by embedding the microsolvated cluster in C-PCM bulk solvent, obtaining converged redox potential results with fewer solvation shells. This combined implicit−explicit solvent model, together with GPU-accelerated QM methods, enabled rapid generation of a large data set of explicit-solvent-calculated redox potentials for 165 redox couples, allowing us to investigate the source of errors in explicit solvent redox potential calculations.

2. THEORY

The general formula for redox potential calculations is given by the Nernst equation

$$E^\circ = -\frac{\Delta G^{\mathrm{EA}}_{(\mathrm{sol})}}{n_e F} - E^\circ(\mathrm{REF}) \tag{1}$$

where $\Delta G^{\mathrm{EA}}_{(\mathrm{sol})}$ is the free energy change associated with reduction at standard conditions in the solution phase, n_e is the number of electrons, and F is the Faraday constant. In this work, the one-electron ferrocenium/ferrocene (Fc^+/Fc) couple is used as the reference, because Fc^+/Fc is the internal reference used in the experimental measurement of the redox potential of the ROP313 data set[26] and is known to be useful to reduce both experimental[27,28] and computational errors,[29,30] especially when varying the solvent in which the redox potential is measured.

2.1. Implicit Solvent Calculation of the Redox Potential. In the implicit solvent models, $\Delta G^{\mathrm{EA}}_{(\mathrm{sol})}$ is calculated based on the Born−Haber cycle,[21] given by

$$\Delta G^{\mathrm{EA}}_{(\mathrm{sol})} = G_{\mathrm{PCM}}(\mathrm{red}) - G_{\mathrm{PCM}}(\mathrm{ox}) + \Delta H^{\mathrm{T}} - T\Delta S(g) \tag{2}$$

Here, $G_{\mathrm{PCM}}(\mathrm{red})$ and $G_{\mathrm{PCM}}(\mathrm{ox})$ are the free energy of the reduced and oxidized species obtained from PCM calculations including electron energy and solvent−solute interaction energy, whereas ΔH^{T} and $-TS(g)$ are the gas-phase enthalpy and entropy contributions to the Gibbs free energy. Recent studies on large data sets show that the gas-phase enthalpy and entropy contributions have very limited influence on the accuracy of redox potential prediction and can be omitted to avoid computational costs related to vibrational frequency calculation.[4] Hence, the equation for redox potential calculations in implicit solvent in this work is simplified as

$$\Delta G^{\mathrm{EA}}_{(\mathrm{sol})} = G_{\mathrm{PCM}}(\mathrm{red}) - G_{\mathrm{PCM}}(\mathrm{ox}) \tag{3}$$

2.2. Explicit Solvent Calculation of the Redox Potential. The redox potential of the explicitly solvated molecules is calculated with the thermodynamic integration (TI) method.[31] Specifically, we use the linear response (LR) approximation of TI[32−34] to avoid simulations of the nonphysical superposition state of the reduced and oxidized forms of the system.[19] For each redox couple, two QM/MM molecular dynamics simulations are performed separately for the reduced and oxidized states. The free energy difference of the reduced and oxidized states is evaluated by thermally averaging the vertical energy gap of the reduced and oxidized states (eq 4).

$$\Delta G_{\mathrm{TI}} = \frac{1}{2}(\langle G_{\mathrm{red}} - G_{\mathrm{ox}}\rangle_{\mathrm{ox}} + \langle G_{\mathrm{red}} - G_{\mathrm{ox}}\rangle_{\mathrm{red}}) \tag{4}$$

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J. Chem. Theory Comput. 2022, 18, 1096−1108

The validity of the TI method and the LR approximation can be estimated by comparing the estimated reorganization energies, λ^{St} and $\lambda^{var.}$, according to the Marcus theory of electron transfer.[35] Here, λ^{St} and $\lambda^{var.}$ are called Stokes reorganization energy and variance reorganization energy, respectively, defined as

$$\lambda^{St} = \frac{\langle \Delta G \rangle_{ox} - \langle \Delta G \rangle_{red}}{2 n_e F} \quad (5)$$

$$\lambda^{var} = \frac{\sigma_{ox}^2 + \sigma_{red}^2}{4 k_b T} \quad (6)$$

where σ is the variance of the vertical energy gap, and k_b is the Boltzmann constant. The reorganization energies λ^{St} and $\lambda^{var.}$ should reach identical values when the LR approximation holds.[34]

3. COMPUTATIONAL DETAILS

3.1. Data Set. We employ the ROP313[26] data set to compare the different computational approaches for predicting the redox potential. This benchmark set with experimental redox potentials for 313 individual molecules is composed of two subsets of 193 organics (OROP) and 120 organometallics (OMROP), respectively. It allows a representative comparison of the accuracy of different computational methods due to the diverse and considerable number of systems. The systems are medium-sized with the number of atoms ranging from 5 to 82 and total electrons ranging from 22 to 427 (Figure 1). The data set contains four different solvents spanning a wide range of dielectric constants, including acetonitrile (MeCN, ϵ = 35.69), water (ϵ = 78.36), dichloromethane (ϵ = 8.93), and dimethylformamide (DMF, ϵ = 37.22). The experimental redox potentials were converted to an internal reference of Fc^+/Fc in an identical solvent.[36,37] The oxidized forms of benchmark molecules have charges ranging from −4 to 2. The organometallics possess a variety of third row (Ti, V, Cr, Mn, Fe, Ni, Co) and fourth row (Ru, Rh, Os, Ir) transition metal centers. The accuracy of redox potential calculations for OROP and OMROP is analyzed separately due to the typically larger errors in the calculation of organometallic systems.[26]

3.2. DFT Calculations. All geometry optimizations and single point energy calculations were performed with the GPU-accelerated quantum chemistry package TeraChem.[38] Unless otherwise specified, the density functional theory (DFT) calculations use the B3LYP hybrid functional with the DFT-D3 empirical dispersion correction,[39] combined with either the 6-31G* basis set or LANL2DZ[40] effective core potentials for the transition metals and I and Br. To test the robustness of our machine learning models for different functionals, a set of hybrid (B3LYP-D3, PBE0-D3) and range-corrected hybrid (ωb97-D3, ωb97X-D3, ωPBEh-D3, CAM-B3LYP-D3) functionals commonly used for redox potential calculations is also employed to calculate single point energies to obtain redox potentials. The geometries optimized with B3LYP-D3 are used in all cases.

For implicit solvent model calculations, geometry optimizations were carried out with the TRIC[41] optimizer using default tolerances of 4.5×10^{-4} hartree/bohr for the maximum gradient and 1×10^{-6} hartree for the change in self-consistent field (SCF) energy between steps. Initial structures, charge, and spin states were obtained from the ROP313 data set originally optimized with ωB97x-c3. Level-shifting[42] values of 0.3 H_a for virtual orbitals were applied if the calculation did not converge without level-shifting. Solvation energies were obtained from single point energies with a conductor-like polarizable continuum model (C-PCM)[43,44] as implemented in TeraChem. The solute cavity was built using defaults available for nonmetals in TeraChem (i.e., 1.2× Bondi's van der Waals radii[45]), and we provided standard van der Waals radii[46] for metals, which were also scaled by 1.2.

3.3. Explicit Solvent Calculations. Due to the significantly higher computational costs of the explicit solvation calculations, only 165 organic redox couples solvated in MeCN are calculated (details in the Supporting Information/SI). Initial explicit solvation configurations are generated with Packmol[47] in a cubic box of side length 56 Å with Na^+ and Cl^- counterions. A multistep approach is chosen to equilibrate the solvated box. First, molecular dynamics parameters are generated for the organic solute molecules with Ambertools[48] using the GAFF force field.[49] The solvent MeCN molecules are described with a customized 6-site model from the literature.[50] The explicitly solvated reference Fc^+/Fc redox couple is set up in a similar approach, but the parameters for the Fc^+/Fc solute are determined from MCPB.py[51] due to the existence of metal−ligand bonds. These nonpolarizable MM force fields allowed long-time dynamics in Amber 20[48] without bond breaking at minimal computational costs. Second, the solvated system is minimized, then slowly heated to 300 K over 20 ps with a Langevin thermostat with a collision frequency of 2 ps^{-1} and a nonbonded cutoff of 8 Å, and pressure equilibrated to 1 bar over 600 ps with a Berendsen barostat with a pressure relaxation time of 1 ps. This long pressure equilibration is chosen to improve the density of the bulk solvent and on the interface between the solute and solvent, which is critical for the explicit redox potential calculation (density as a function of barostat equilibration time available in SI Figure S13). This multistep solvation of each system is automated by AutoSolvate, an in-house python script developed in our group, which is available upon request. The resulting pressure-equilibrated system is the initial structure for the QM/MM[52,53] simulation with TeraChem and Amber 20. The QM region is then solute treated with B3LYP-D3/6-31G*, and the MM region is the explicit solvent. The QM electrostatic cutoff is 8 Å. The QM/MM simulation involves an initial energy minimization (250 steps) to remove potential unexpected bond breaking of the solute due to abruptly switching from the MM to the QM description, followed by 0.5 ps of temperature equilibration with a Langevin thermostat at 298.15 K with a collision frequency of 5 ps^{-1}. The following 5 ps QM/MM trajectory under NVT conditions is used for TI. For each of the two charge states, 200 snapshots are extracted from the 5 ps QM/MM trajectories. The large number of snapshots is required to average out the variance of the vertical energy gap.[54] Each snapshot includes the solute and dozens of solvent molecules. To conduct DFT single point calculations needed by TI on these snapshots with reasonable computational costs, a cutoff needs to be applied to generate a microsolvated cluster that is small enough to calculate with DFT but large enough to generate a converged redox potential. We tested various cutoff values from 2 to 10 Å. We also investigated how the C-PCM implicit solvent applied around these microsolvated clusters impacts the calculated redox potential.

3.4. Machine Learning Models. Machine learning (ML) models are utilized to reduce the errors in computed redox

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potentials compared to experimental values. Due to the relatively small size of the experimental redox potential data set, we trained ML methods on a small set of 9 physics-inspired features related to solvent effects and redox potential calculation, including 4 features for the solute molecule (charge, dipole moment, spin multiplicity, and nuclear repulsion energy of the oxidized state), 4 features for the solvent model (implicit solvation cavity surface area and volume, C-PCM solvation energy, and dielectric constant of the solvent), plus the predicted redox potential (Figure 2). We

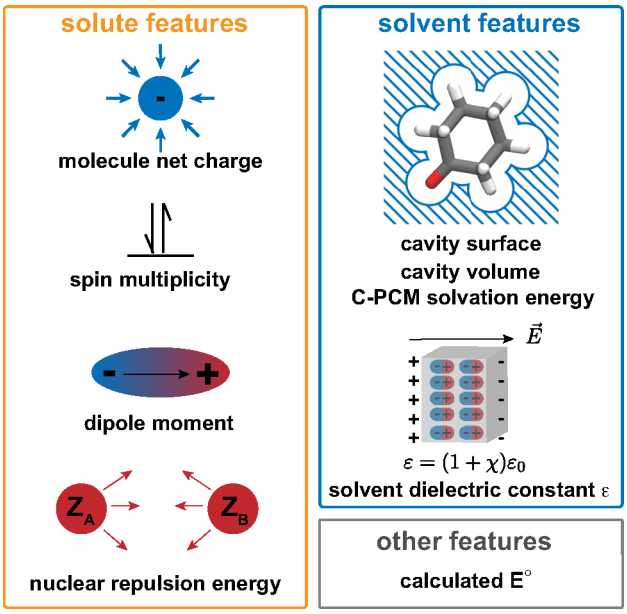


Figure 2. ML features used for training machine learning models to correct errors in redox potential calculations in this study.

compared the performance of different ML models utilizing scikit-learn, including simple linear regression (lin-1), multiple linear regression (lin-m), random forest regression (RF),[55] gradient boost regression (GB),[56] kernel ridge regression (KRR),[57] and artificial neural network (ANN).[58] Hyperparameter optimization for all models was carried out with grid-search cross-validation to prevent overfitting, using a random 80% train/20% test split, with 20% of the training set (16% overall) set aside as the validation subset for hyperparameter selection (SI Text S1). Input features were normalized over the training set to have zero mean and unit variance. Final models with optimal hyperparameters were retrained with the whole training set (80% overall). We repeated the training five times on different 80:20 splittings of the data set, such that the five 20% test sets in total cover all data points. Performance of the ML correction is reported for the combination of the 5 test sets.

3.5. Free Energy Landscape and TICA. The convergence of the conformation dynamics of the QM/MM trajectories for our explicit solvent calculations can be confirmed by analyzing the trajectories. One challenge in analyzing the trajectories is the high-dimensional nature with up to hundreds of individual atoms moving. To analyze the solute conformation changes, we use the dimension reduction method Time-lagged Independent Component Analysis (TICA),[59] which extracts a few dimensions from the high-dimensional raw trajectories. TICA represents the kinetic behavior of the trajectory better than principal component analysis (PCA). The following equations describe how TICA selects the dimensions with the highest kinetic information. The raw Cartesian time-dependent coordinates are converted into rotation- and translation-invariant mean-free features $y_i(t)$, in this case, distances between atoms. For a lag time τ, the covariance matrix $C(0)$ and the time-lagged covariance matrix $C(\tau)$ for the mean-free coordinates are calculated:

$$C_{ij}(\tau) = \langle y_i(t) y_j(t+\tau) \rangle_t \tag{7}$$

$$\mathbf{C}(\tau) v_i = \mathbf{C}(0) v_i \lambda_i \tag{8}$$

The generalized eigenvalue problem allows us to select a few dimensions or eigenvectors v_i with the largest eigenvalues λ_i, which contain most of the kinetic information on this system and have the longest associated time scales. The projection of the raw trajectory on the selected eigenvectors gives a low-dimensional representation of the solute behavior. In this work, only the two slowest dimensions are selected, and the lag time utilized is 5 fs due to the fast conformational dynamics of the systems. In the projection defined by the two selected TICA dimensions, we calculate the free energy, which allows us to visualize both the ensemble of the reached conformations and the transitions between the conformations. The free energy landscape is comparable to the potential energy surface (PES) but with additional degrees of freedom considered.[60] In order to plot the redox potential projection in the TICA dimensions, additional redox potential calculations were performed for conformations uniformly sampled along the two TICA dimensions from the QM/MM trajectories.

4. RESULTS

In the following subsections, we will investigate the errors in the calculated redox potential compared to experimental results and build ML models to correct these errors and reduce computational costs. We will first investigate the accuracy of the implicit solvation redox potential calculations and build ML models to correct the errors related to the imbalanced treatment of different charge states in implicit solvent models. We will also demonstrate that the ML corrections improve the robustness of the calculated redox potentials with respect to DFT functionals. For the explicit solvation approach, we investigate the dependency of the calculated redox potential with respect to the solvent shell size and will demonstrate that by applying C-PCM implicit solvent around the explicit solvent molecules, a much smaller explicit solvent shell is required to obtain converged redox potentials, leading to significantly reduced computational costs. We will then discuss the impacts of applying the ML correction to the explicit-solvent-calculated results and the source of errors in the explicit solvation redox potential calculations. Comparisons between our approach and other approaches will be discussed at the end.

4.1. Errors in Implicit Solvent Redox Potential Calculations. It is known that redox potential calculations in implicit solvent typically have relatively large uncertainties in the solvation free energy since electrochemical half-reactions involve the consumption or generation of charge species.[6] Although low unsigned errors of less than 100 mV were observed in small batch studies, more realistic errors can be much larger when more diverse and larger test sets are considered.[26,61] In our calculation of the ROP313 data set with

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B3LYP-D3 and C-PCM (Figures 3 and S3), two types of errors are present: systematic bias and large-error outliers.

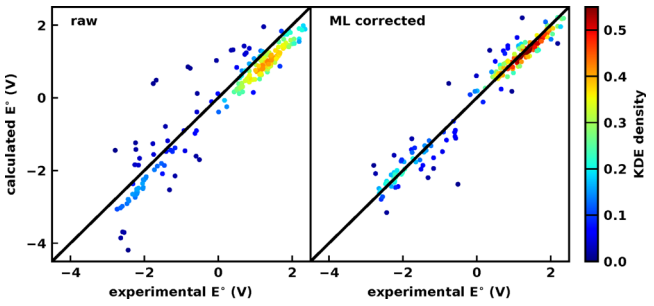


Figure 3. Parity plots of the implicit solvent approach calculated vs experimental redox potentials for compounds in the OROP data set. Data points are colored by kernel density estimation (KDE) density values, as indicated by the color bar on the right. (left) Raw data from implicit solvation calculations. (right) ML corrected data obtained with the best performing KRR model.

Compared to experimental results, the calculated redox potentials of the OROP data set show a systematic underestimation, especially in the higher value range with high data density (Figure 3). This type of systematic bias is commonly seen in implicit solvent redox potential calculations and is often corrected with simple linear fitting between the calculated and experimental redox potentials.[21,62] After applying the traditional simple linear regression correction for OROP, the mean absolute error (MAE) is reduced from 0.43 to 0.33 V (Table 1).

However, the traditional linear fitting correction typically cannot help fixing errors that do not follow a systematic linear trend, including big-error outliers. For example, transition metal complexes with excess positive/negative charges are known to be typical outliers because implicit solvent models usually have an imbalanced treatment of solvation free energy for molecules with different net charges.[16,17] Inspired by this idea, we analyzed the errors of redox couples of different charge states. For the OROP data set, most systems have either +1/0 or 0/−1 charge states. Statistics on the distribution of signed errors of these two groups have different trends (Figure 4). The mean signed errors (MSEs) of both groups are negative, but the MSE of the +1/0 group (−0.25 V) is further from zero than that of the 0/−1 group (−0.08 V). The difference in MAE is even more significant: the 0/−1 group

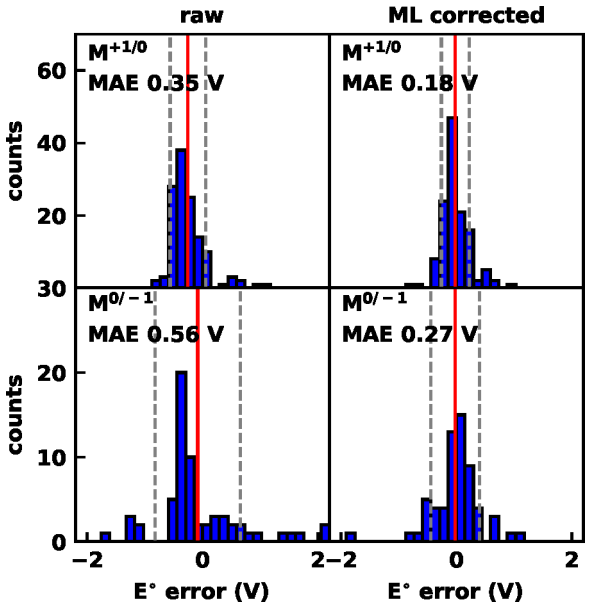


Figure 4. Histogram of errors (in V) in implicit solvent approach calculated redox potentials for two subsets of the OROP data set with different charges. (top) Histograms for the subset with the oxidized/reduced charge state of +1/0 before and after the ML correction (KRR model). (bottom) Histograms for the subset with a charge state of 0/−1. The red vertical lines indicate the mean signed error (MSE) for each subset. The gray dashed vertical lines show the size of the standard deviation.

has a much higher MAE of 0.56 V than the 0.35 V of the +1/0 group due to a larger numbers of outliers. Similar results are observed for the OMROP data set (SI Figure S5).

Because of different distributions of errors in systems with different charges, simple linear fitting cannot effectively correct the errors. A few methods have been proposed in previous works to fix these charge-dependent errors, including the pseudocounterion solvation (PCIS) scheme that applies a charge-dependent correction formula[16] and the variable-temperature H atom addition/abstraction approach that modifies the thermodynamic cycle.[17] Here, we correct the systematic bias and charge-dependent outliers simultaneously by training machine learning models that are aware of the complex mapping between solute/solvent features and the errors in implicit solvent calculations, which will be discussed further in Section 4.2.

Table 1. DFT Functional-Dependence of the Test Set MAE (in V) of Redox Potentials Predicted by the Implicit Solvent Model in Combination with Various ML Correction Models for the OROP Data Set[a]

MAE (V)	implicit	lin-1	lin-m	KRR	GB	RF	ANN	best ML
B3LYP-D3	0.43	0.33	0.32	0.22	0.27	0.27	0.25	KRR/0.22
B3LYP	0.41	0.34	0.32	0.23	0.27	0.26	0.27	KRR/0.23
ωb97-D3	0.64	0.55	0.36	0.33	0.32	0.30	0.33	RF/0.30
ωb97X-D3	0.47	0.48	0.35	0.29	0.33	0.27	0.30	RF/0.27
ωPBEh-D3	0.44	0.44	0.34	0.26	0.33	0.27	0.30	KRR/0.26
PBE0-D3	0.30	0.31	0.31	0.21	0.25	0.25	0.27	KRR/0.21
CAM-B3LYP-D3	0.40	0.42	0.34	0.26	0.31	0.28	0.25	ANN/0.25
functional sensitivity	0.34	0.23	0.05	0.11	0.08	0.05	0.08	RF/0.05
best functional	0.30	0.31	0.31	0.21	0.25	0.25	0.25	KRR/0.21

[a]For each functional, the best ML correction model with the smallest MAE (in V) is shown in the last column. For each column (ML model), functional sensitivity denotes the MAE difference between the best and worst performing functionals, and best functional denotes the MAE of the best performing functional.

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J. Chem. Theory Comput. 2022, 18, 1096−1108

Table 2. DFT Functional-Dependence of the Test Set MAE (in V) of Redox Potentials Predicted by the Implicit Solvent Model in Combination with Various ML Correction Models for the OMROP Data Set[a]

MAE (V)	implicit	lin-1	lin-m	KRR	GB	RF	ANN	best ML
B3LYP-D3	0.76	0.44	0.46	0.46	0.50	0.47	0.46	lin-1/0.44
B3LYP	0.82	0.46	0.47	0.47	0.50	0.45	0.47	RF/0.45
ωb97-D3	1.35	0.67	0.70	0.75	0.79	0.74	0.73	lin-1/0.67
ωb97X-D3	1.27	0.60	0.63	0.60	0.71	0.67	0.77	KRR/0.60
ωPBEh-D3	1.10	0.58	0.60	0.61	0.62	0.60	0.64	lin-1/0.58
PBE0-D3	0.83	0.45	0.47	0.45	0.47	0.43	0.48	RF/0.43
CAM-B3LYP-D3	1.32	0.59	0.61	0.63	0.72	0.67	0.65	lin-1/0.59
functional sensitivity	0.59	0.23	0.25	0.30	0.33	0.32	0.31	lin-1/0.23
best functional	0.76	0.44	0.46	0.45	0.47	0.43	0.46	RF/0.43

[a]The "implicit" column shows the implicit solvation results without any ML correction. For each functional, the best ML correction model with the smallest MAE (in V) is shown in the last column. For each column (ML model), functional sensitivity denotes the MAE difference between the best and worst performing functionals, and best functional denotes the MAE of the best performing functional.

4.2. ML Corrections for Implicit Solvent Calculations. We train ML models to correct the errors in implicit solvent redox potential calculations, focusing on reducing the uncertainty in solvation free energy. Since the C-PCM solvation free energy is determined by the charge distribution of the solute, the solvent dielectric constant, and the cavity shape, we use features that represent these physical properties and can be easily extracted from calculation output files (Figure 2). Specifically, net charge, dipole moment, and spin multiplicity of the solute represent the solute charge distribution; solvent dielectric constant describes the electrostatic screening strength; cavity surface area and volume describe the cavity shape; nuclear repulsion energy is closely related to the widely used molecule descriptor, Coulomb matrix,[63] and contains implicit information about the solute size and structure (detailed discussion available in SI Text S3, Figures S9 and S12); and the calculated redox potential is also included in the features to allow for various corrections in different ranges of the redox potential.

For the B3LYP-D3 results discussed in Section 4.1, we observed reduced MAE after corrections with all types of ML models (Tables 1 and 2). For the OROP data set, all methods have significantly reduced the MAE by 23%−49%, with KRR as the best performing model reducing the MAE from 0.43 to 0.22 V. Even more prominent improvement is observed in the OMROP data set, where all ML models present an MAE reduction of over 34%, with lin-1 as the best performing model reducing the MAE from 0.76 to 0.44 V.

Further analysis shows that the error reduction is achieved by fixing both the systematic bias and the outliers discussed in Section 4.1. Compared to the raw calculated redox potential for OROP, the ML-corrected results distribute more evenly on both sides of the diagonal of the parity plot (Figure 3), significantly removing the overestimation trend. More importantly, the more challenging charge-state dependent errors are reduced simultaneously (Figure 4). The MSEs of both the 0/−1 group and the +1/0 group are both closer to 0 after applying the ML correction (Figure 4). The MAEs of both groups are reduced by about 50%, demonstrating the balanced correction for differently charged groups. Mitigation of large-error outliers is even more obviously shown by the reduced number of systems with an error above 1 V. After applying the ML model, such systems are reduced from 14 to 2 for the OROP data set and from 35 to 11 for the OMROP data set.

4.3. Robustness of the ML Correction to the DFT Functional Choice. In this section, we demonstrate the robustness of the ML corrections developed in Section 4.2 with respect to the choice of the DFT exchange correlation (XC) functional. It is widely known that the accuracy of redox potential calculation in the implicit solvent is highly sensitive to the choice of the XC functional, and the optimal functional is usually system dependent, leading to the lack of general strategy to obtain highly accurate computational redox potentials for large, diverse data sets.[26] This high sensitivity is also observed in our calculations. For the OROP data set, the MAEs of the best performing PBE0-D3 (0.30 V) and the worst performing ωb97-D3 (0.64 V) differ by 0.34 V (Table 1). An even larger variation (0.76 V−1.35 V) in the MAEs of different functionals is observed for the OMROP set (Table 2).

In contrast, ML corrected results are always less sensitive to the choice of the DFT functional. For the OROP data set, any of the nonlinear ML models can reduce the MAEs of each functional and reduce the performance difference among functionals to no more than 0.11 V (Table 1). KRR is generally the most efficient ML model for OROP, reducing the organic MAE down to 0.21 V. With the KRR correction, the MAE of each functional is reduced by at least 0.08 V, and a greater improvement is observed for the originally worse performing functionals. As a result, the performance difference between any two functionals is reduced to less than 0.11 V from 0.34 V (Figure 5). A similar reduction of functional sensitivity after the ML correction is observed for the OMROP data set (Table 2). However, due to the smaller OMROP data set size, the linear ML correction performs similarly to nonlinear ML corrections. Overall, RF has the best performance, reducing the OMROP MAE down to 0.43 V, whereas the lowest sensitivity to function choice [MAE(worst) − MAE(best) = 0.23 V] is observed with lin-1. Although the functional sensitivity is much smaller after the ML correction, B3LYP-D3 and PBE0-D3 are overall the best for both OROP and OMROP data sets. Therefore, B3LYP-D3 will be used in the explicit solvent model calculations in Sections 4.4 and 4.5.

The effectiveness of the ML corrections in reducing MAE is also verified for DFT calculations with different basis set choices (SI Figures S2, S4, and S10). Regardless of the basis set used, the ML corrections reduce the MAE by a similar extent. Without the ML correction, we observe an increased MAE for OMROP when larger basis sets with diffuse functions (6-31+G*, 6-31++G**) are used, consistent with previous reports of this counterintuitive trend in redox potential

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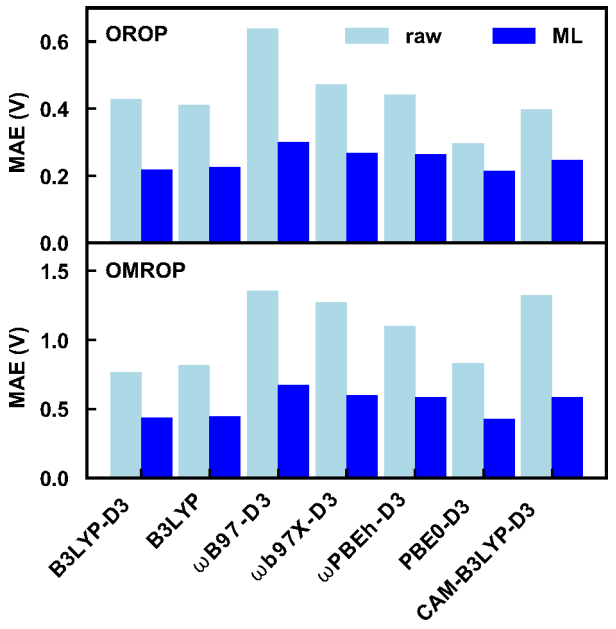


Figure 5. Comparison of the MAE for redox potential prediction before and after ML corrections for different DFT functionals. For each functional, the best performing ML model's result is reported. Results for OROP and OMROP data sets are shown in the top and bottom panels, respectively.

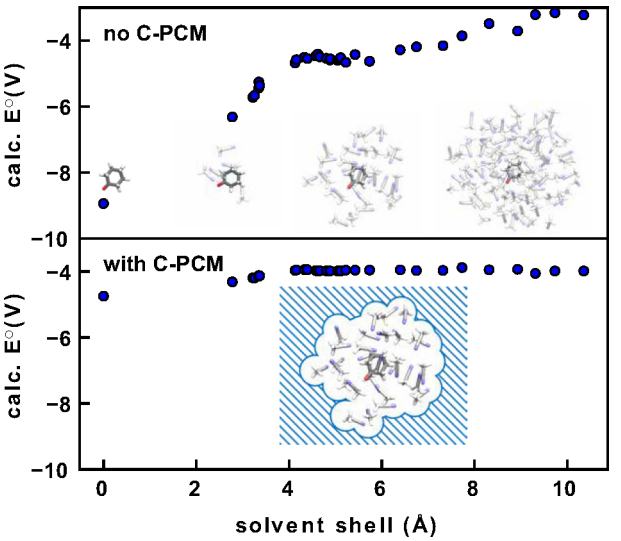


Figure 6. Convergence of the redox potential with respect to the radius of the explicit solvation shell for cyclohexanone in the MeCN solution. (top) Results obtained with the explicit solvation shell only. Structures of microsolvated clusters of different sizes are shown in the inset. (bottom) Results obtained with the explicit solvation shell embedded C-PCM. Schema of the combined explicit−implicit solvation is shown in the inset.

calculations for transition metal complexes.[64] After the ML correction, calculations obtained with larger basis sets show higher accuracy for both OROP and OMROP (Figures S2, S4, and S10), but the improvement relative to 6-31G* is only up to 0.04 V.

4.4. Convergence of Explicit Solvent Redox Potential Calculations. Explicit solvent models are expected to provide a more accurate prediction of the redox potential, but their application is limited by the higher computational costs compared to implicit solvent calculations. It was shown in previous studies that multiple layers of solvent shells are needed to obtain the converged redox potential, requiring QM calculations of large microsolvated systems with hundreds of atoms.[22,23,65] Including long-range polarization[22] or dielectric continuum[66] can improve the accuracy. Here, we demonstrate that embedding the microsolvated cluster in C-PCM can effectively reduce the number of explicit solvent molecules needed without affecting the accuracy.

We first investigated the dependence of calculated redox potential values on the solvation shell size for system 141, cyclohexanone, picked from the ROP313 data set (17 atoms, 54 electrons, Figure 6). For the traditional explicit solvent model without C-PCM, the redox potential varies strongly with solvent shell size and only converges above 10 Å (Figure 6). The existence of multiple plateaus at around 5, 8, and 10 Å is caused by the boundaries of solvent layers (analysis of solvent layers available in SI Figure S14). In contrast, the redox potential obtained with the microsolvated cluster embedded in C-PCM converges rapidly as the solvent shell size increases and remains unchanged after 4 Å (Figure 6). The redox potential convergence at 4 Å coincides with the location of the gap between the first and the second solvent layers, indicating that only the first solvent layer has to be treated explicitly for microsolvated clusters embedded in C-PCM. The faster convergence of the explicit solvent model embedded in C-PCM can be explained by a more efficient representation of long-range and polarization interactions with the bulk solvent compared to including only a few explicit solvation shells. The explicit solvation redox potentials calculated with and without C-PCM converge to different values, which is likely to be caused by the inherent deficiencies of the C-PCM model in describing solvation free energy. We will show that these potential artifacts caused by C-PCM can be effectively removed with ML corrections, as demonstrated with the implicit solvent approach in Sections 4.2 and 4.3.

To demonstrate that the fast convergence holds for different systems, we repeat the investigation to determine the optimal solvent shell size with and without the C-PCM embedding for two additional OROP systems (SI Text S2). For each system, we calculate the absolute deviation of the redox potential at a given solvent shell size compared to the reference value obtained at 10 Å. The averaged absolute deviations over the 3 systems at solvent shell sizes 2, 4, 6, and 8 Å are 5.7, 2.5, 2.3, and 1.8 V, respectively, for the explicit solvent model without C-PCM. In contrast, these values are 0.65, 0.03, 0.05, and 0.03 V for the explicit solvent model embedded in C-PCM. In summary, the explicit solvent model without C-PCM does not converge even at 8 Å, whereas a 4 Å solvent shell embedded in C-PCM reaches an accuracy and convergence sufficient for most use cases. The embedding of microsolvated clusters in C-PCM reduces the required solvation shell size and computational cost while maintaining accuracy. Unless otherwise specified, explicit solvation refers to embedding of microsolvated clusters in C-PCM.

4.5. Accuracy of Explicit Solvent Calculations and ML Corrections. Thanks to the reduced computational costs enabled by C-PCM embedding, we are able to rapidly curate a data set of redox potentials of 165 organic systems calculated with an explicit solvent model. Compared to previous studies with explicit solvent models typically applied on only a few systems,[67] this larger data set allows us to summarize some

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J. Chem. Theory Comput. 2022, 18, 1096−1108

statistically meaningful trends in this type of calculation. Similar to implicit solvent calculations, both systematic biases and large-error outliers are present in our explicit solvent calculations (Figure 7). The predicted values are systematically

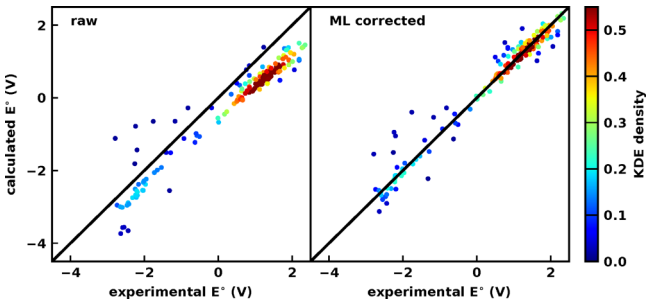


Figure 7. Accuracy of the predicted redox potentials with the explicit solvent approach with the microsolvated cluster embedded in C-PCM for 165 explicitly solvated OROP systems. (left) Raw results without the ML correction show a systematic bias. (right) Results after the ML correction with RF.

below the experimental values, and in the low redox potential region ($E^\circ \leq 0$ V), there are more outliers with the overestimated redox potential, similar to the trend in implicit solvent calculations (Figure 3).

We then train machine learning models to correct the systematic bias and outliers in the explicit solvent redox potential calculations, using the same procedure as the implicit solvent ML corrections (Figure 7). All tested ML models significantly reduce the test set MAE from 0.64 V to no more than 0.24 V, with the best performance (0.19 V) observed for RF (Table 3). The number of outliers stays almost unchanged after applying all ML corrections.

We further compare and contrast the performance of the explicit and implicit solvent approaches on the same subset of 165 organic systems (Table 3, SI Figure S11). Although previous small batch studies indicate that the computationally more expensive explicit solvent approach tends to be more accurate than the implicit solvent approach, our data show exceptions. Before the ML correction, the explicit solvent approach has a larger MAE (0.64 V) than the implicit solvent approach (0.40 V) due to a larger systematic bias. However, it produces fewer outliers (3 vs 11), which can be attributed to the more accurate representation of solvent−solute interactions in explicit solvation. After the ML correction, the explicit solvent approach has a significantly improved MAE (0.19 V) similar to the implicit solvent approach (0.20 V), but its advantage in generating fewer outliers disappears due to the significant reduction of outliers for the implicit solvent approach (Table 3). These counterintuitive observations motivate us to further investigate the errors in our explicit solvent redox potential calculations in Section 4.7.

4.6. Analysis of Error Sources in Explicit Solvent Calculations. To understand why the explicit solvent approach did not outperform the implicit solvent model as expected, we further analyze potential sources of errors and which systems are more likely to suffer from these errors. A typical error source for the explicit solvent model is the sampling of redox couple configurations, so we picked a representative system (ROP313 system 29, 1-bromo-4-prop-1-enylbenzene) and investigated the variation of its calculated redox potential projected onto a TICA dimension-reduced landscape of the solute conformations (Figure 8). The

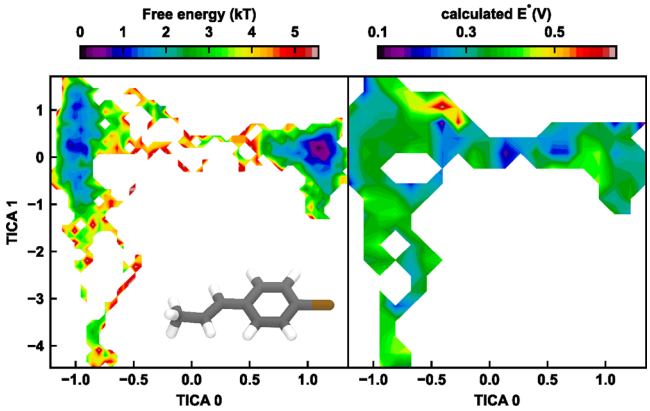


Figure 8. Dimension reduced landscape for ROP313 system 29, 1-bromo-4-prop-1-enylbenzene. (left) Free energy landscape along the slowest two TICA dimensions shows three minima. (right) The averaged redox potential along the slowest two TICA dimensions. The redox potential varies significantly across the conformational space.

projected redox potential is estimated only for the oxidized charge state since the free energy landscapes for the two charge states are generally different and separate. The dimension-reduced landscape is generated from the QM/MM trajectory used for TI calculation of the redox potential, as described in Section 3.5. The free energy landscape of system 29 has three local free energy minima, with fast transition times in the subpicosecond range (Figure 8, left). The sparse sampling of the transition regions indicates that longer QM/MM trajectories are necessary to converge the sampling. Although conformations around the local minima have similar redox potential values of about 0.4 V, the transition regions between the minima have some conformations with redox potential values as high as 0.7 V or as low as 0.1 V (Figure 8, right). Similar analysis on two additional systems from the OROP data set showing different numbers of local minima on the free energy landscape is available in SI Figures S6 and S7. For systems with multiple local minima, insufficient sampling or analyzing only one conformation as in the implicit solvation

Table 3. Performance of the Explicit Solvent Approach before and after Various ML Corrections[a]

	explicit	lin-m	KRR	GB	RF	ANN	best ML	implicit	implicit+ML
MAE (V)	0.64	0.22	0.20	0.22	0.19	0.24	RF/0.19	0.40	0.20
errors > 1 V	3	3	2	3	4	4	KRR/2	11	1

[a]The mean absolute error (MAE) in V and the number of systems with an error above 1 V are shown for the analyzed 165 systems from OROP. The "explicit" column shows the explicit solvation results without any ML correction. The ML method columns and best ML column show the explicit solvation results with ML corrections. The implicit and implicit+ML columns show the results of the implicit solvent approach without the ML correction and with the best performing model (KRR). All calculations are performed with B3LYP-D3/6-31G*.

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J. Chem. Theory Comput. 2022, 18, 1096−1108

approach can be one of the contributions to the observed redox potential errors and outliers.

Another potential error source is the LR approximation utilized in TI for the explicit solvent redox potential calculation. We compare the estimated reorganization energies λ^{St} and $\lambda^{var.}$ to validate the LR approximation. For the 165 explicitly solvated systems calculated, the median λ^{St} is 0.37 V, and the median $\lambda^{var.}$ is 0.30 V. The small difference between the two estimates indicates that the LR approximation is generally valid for the investigated systems. Histograms of the reorganization energies show that most systems have λ^{St} and $\lambda^{var.}$ below 0.5 V (SI Figure S8), similar to reported values of previous small batch studies.[34] However, some outlier systems have λ^{St} or $\lambda^{var.}$ up to around 2.5 V, comparable to literature values for phenol.[66] For most systems, the λ^{St} and $\lambda^{var.}$ show similar values, but $\lambda^{var.}$ is systematically smaller than λ^{St} by around 0.04 V. We also observe a few exceptions where the λ^{St} and $\lambda^{var.}$ differ by more than 0.5 V, possibly caused by insufficient sampling. Differences between $\lambda^{var.}$ and λ^{St} have been previously reported for metalloproteins,[32,68] with non-parabolic shapes of the free energy surfaces,[68] polarizability,[69] nonergodic effects,[70] and inner-sphere/solute or outer-sphere/solvent effects[71] discussed as causes.

Furthermore, we observe a correlation between the size of reorganization energy and the size of redox potential errors: systems with higher reorganization energies tend to have higher redox potential errors. The ML corrected MAE for systems with $\lambda^{St} > 0.5$ V is 0.38 V, significantly higher than the 0.15 V MAE for systems where $\lambda^{St} < 0.5$ V. Alternatively, using $\lambda^{var.}$ as the criterion gives consistent results (SI Table S1).

Finally, we examine the counterions as a possible source of error. The counterions are originally added for neutralizing the charge of the system during MM dynamics in the periodic boundary condition. For most systems, the counterions stayed far from the solute and were not included in the microsolvated clusters used for TI. However, for some solutes with higher net charges, the counterions may migrate to be close to the solute and be included in the TI snapshots. Although we suspect the electrostatic interaction between the counterions and the solute may cause artifacts in the calculated redox potential, we see only a small increase of MAE when counterions are present in the microsolvated clusters (SI Table S1).

4.7. Comparison with Other Approaches. To better understand the advantages and limitations of our approach, we compare the performance of our approach with some other works focusing on the accuracy of redox potential prediction relative to experiments. Specifically, we make comparisons from the following aspects: (1) accuracy of our implicit solvent approach compared to other works on diverse data sets; (2) accuracy of our explicit solvent approach compared to other works on diverse data sets; (3) use of ML in redox potential prediction; and (4) consideration of other experimental conditions.

For the accuracy of the implicit solvent approach, the most direct comparison is with the original benchmark calculations of ROP313 by Grimme and co-workers.[26] Without the ML correction, the MAEs of our calculations with the best performing functional are 0.30 V for OROP and 0.76 V for OMROP, higher than the 0.25 V for OROP (B97-c) and 0.33 V for OMROP (B97-D3) reported in the Grimme work. This higher MAE is expected because the MAEs of Grimme's work were reported for a pruned subset where 10 out of 192 OROP systems and 54 out of 120 OMROP systems were excluded because of large errors >4 V or not converged SCF. In contrast, we converged all calculations and did not exclude any outliers when calculating the MAEs. After applying the ML correction, our MAEs (0.21 V for OROP, 0.43 V for OMROP) are similar to Grimme's results despite including all outliers. Another representative work about benchmarking implicit solvent approaches against an experimental data set is by Truhlar and co-workers.[72] The optimal G3(MP2,CC)(+) method was combined with four variants of C-PCM models to benchmark the reduction potential calculation of 53 organic molecules (nitroxides, phenols, and amines) in aqueous solutions. The overall MAEs range from 0.19 to 0.23 V, in the same order as our OROP results before the ML correction. However, it is worth noting that Truhlar's benchmark used standard hydrogen electrode (SHE) as the reference, whereas we used Fc^+/Fc that tends to have systematic error cancellation effects.[73] Further investigation in the performance differences between SHE and Fc^+/Fc references will allow a better comparison of the results. McNeill and co-workers recently benchmarked another chemically diverse organic data set,[61] reaching an overall MAE of 0.4 V after a linear regression correction. Their work also included some two-electron transfer processes, which are more complicated than the one-electron transfer processes investigated in this work.

For the thermodynamic integration approach, most of its applications focus on organometallics that are difficult to predict with implicit solvent approaches.[19,34,67,74] However, our explicit solvent calculations are for organic molecules due to the relative ease of automating the MD simulations needed to curate a large data set. To make the performance comparison straightforward, we focus on two works about organic redox couples. Sprik and co-workers benchmarked the prediction for a set of 12 organic redox-couples, with MAEs of 0.51 V for the GGA functional, BLYP, and 0.26 V for the range-corrected hybrid functional, HSE06. Before the ML correction, our overall MAE (0.64 V) is higher, but we have a much larger data set of 165 organic redox couples. After the ML correction, our overall MAE (0.19 V) is lower than Sprik's. However, most molecules in our calculations are solvated in the aprotic solvent, MeCN, which has weaker solvent−solute interaction than the aqueous solvent studied in Sprik's work. Future application of our protocol to aqueous solutions and organometallics will help us better understand the performance. Bjornsson and co-workers proposed a multistep explicit solvation protocol to calculate the redox potential with the Drude[75] polarized QM/MM scheme and reached a small MAE of 0.13 V for 19 phenol systems.[22] Our work has a slightly higher MAE of 0.19 V even after the ML correction, both because of the significantly larger and more diverse data set and the nonpolarizable MM force field used for QM/MM sampling. However, the required explicit solvent shell radius of 5 Å for Bjornsson's work is comparable to the 4 Å in our work, regardless of the differences in the protocols.

For the use of ML in redox potential prediction, most works have focused on training ML models to reproduce the QM calculated redox potential without direct comparison to experimental results.[76−78] For example, Allam and co-workers[77] compared the performance of different nonlinear ML methods (KRR, GB, and ANN) for reproducing DFT calculated redox potentials of organic electrodes. Despite the similar set of ML methods as used in our work, the purpose and the context of applying these ML models are different. To link calculated redox potentials to experiments, Aspuru-Guzik

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J. Chem. Theory Comput. 2022, 18, 1096−1108

and co-workers calibrated semiempirical quantum chemistry calculations in implicit solvent against available experimental data with Gaussian process (GP) regression, which was then applied to the calculated redox potentials of a large set of non-natural biomedical redox reactions.[5]

In addition to the solvent, pH is also a critical environment parameter impacting redox potential because the one-electron reduction of some species (e.g., amines and phenols) also involves changes in the protonation state. A recent work by de Silva and co-workers[79] proposed a protocol combining chemical informatics and quantum chemistry to calculate the pH-dependent redox potential. The workflow first determines the most favorable protonation state at pH = 0 based on the SMILE string and then searches for the most stable conformer at the MM level. The redox potential is first calculated for the pH = 0 conformer and then converted to other pH values. Due to the limited number of experimental pH-dependent data points available to training ML models, our approach does not consider the effects of pH at the moment.

5. CONCLUSION

This work exploits machine learning to reduce the errors relative to experimental measurements in redox potential calculations in both implicit and explicit solvents.

For the implicit solvent approach, we apply ML corrections to mitigate the systematic biases and reduce the number of outliers. This approach enabled us to reach an MAE of 0.21 V for the OROP data set and 0.43 V for the OMROP data set without any systems excluded, demonstrating improved accuracy compared to previously reported calculations on the same data sets without the ML correction.[26] More importantly, the ML correction decreases the sensitivity of predicted redox potential values to functional choice, a long-standing challenge affecting the redox potential prediction accuracy for large, diverse data sets. With more experimental redox potential data points reported and larger experimental data sets available in the future, our ML corrections are expected to achieve even better performance through model retraining.

For the explicit solvent approach, we embedded the microsolvated clusters in C-PCM to reduce the computational costs. The redox potential calculated with the combined implicit−explicit model converges with a 4 Å solvent shell, significantly smaller than around the 10 Å shell required by the explicit-solvent-only counterpart. The resulting acceleration allowed rapid explicit solvent redox potential calculation for 165 OROP compounds, with a smaller number of outliers than implicit solvent calculations on the same set of systems. We then adapted the aforementioned ML correction to these combined implicit−explicit calculations to reduce potential artifacts brought in by C-PCM and obtained a very similar performance compared to the implicit solvent approach after the ML correction.

Finally, we analyzed the potential error sources of the explicit solvent approach. Dimension reduction analysis of explicitly solvated trajectories indicates a primary source of errors to be the insufficient thermodynamic sampling of solute conformations. Although the explicit solvent approach overcomes the drawback of the implicit solvent approach that considers only one optimized geometry, solutes with multiple minima may not have been sufficiently sampled due to limited trajectory lengths. The second error source is the limited validity of the LR approximation made in our TI protocol, as revealed by the discrepancy between the reorganization energies λ^{St} and $\lambda^{var.}$ in our data set. We observed that counterions are not a large error source since only a small difference in accuracy was observed between the groups with and without counterions.

We expect that the ML-based correction strategies proposed in this work will enable rapid curation of computational redox potential data sets with an improved agreement with experimental measurements. The increased robustness to QM method choice will allow automated workflows for redox potential calculation without postadjusted, system-specific calculation parameters. These will enhance the accuracy and efficiency of high-throughput computational design and discovery in catalysis and energy storage.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.1c01040.

Additional implicit redox potential plots; additional redox potential projections; reorganization energies, ML results; hyperparameter ranges; and description of ZIP file (PDF)

Geometry optimized xyz coordinates for all structures; raw and ML corrected redox potential predictions for both implicit and explicit solvation; and utilized system subsets, trained ML models, and input feature data (ZIP)

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Notes

The authors declare no competing financial interest.

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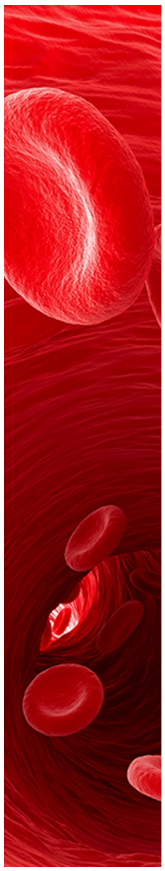
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